Fabrication and Integration Process

1. Material Choice Justification :

The foundation of the vehicle consists of a laser-cut MDF box that fully encloses the main circuit board and wiring. MDF was specifically chosen for this layer because it is a dense, rigid material that provides structural stability while remaining lightweight enough not to overload the motors. By housing the circuit inside this box, the electronics are fully protected from external factors and the opaque nature of MDF also keeps the wiring hidden from the user, giving the vehicle a clean appearance.

The laser-cut acrylic box acts as the delivery compartment where items can be placed and transported by the vehicle. Aesthetically, acrylic gives a smooth, glossy finish to the vehicle. It has a modern and visually appealing look that contrasts well with the solid MDF base beneath it. It is also a rigid and durable material, meaning it can bear the weight of delivered items without warping or cracking.

The MDF under the DC motors acts as a base for the motors, distributing the combined weight of both boxes evenly across the wheel assembly. This prevents the motors from being overloaded or misaligned under the vehicle's total weight, which could otherwise cause uneven movement.

2. Fabrication Steps :

1. Designed digital files for both the MDF and acrylic boxes using, with precise measurements tailored to fit the circuit, wiring, and delivery compartment requirements.

2. Submitted the designs for approval, made the necessary revisions based on feedback, and procured the MDF and acrylic sheets in the appropriate sizes and thickness once the designs were

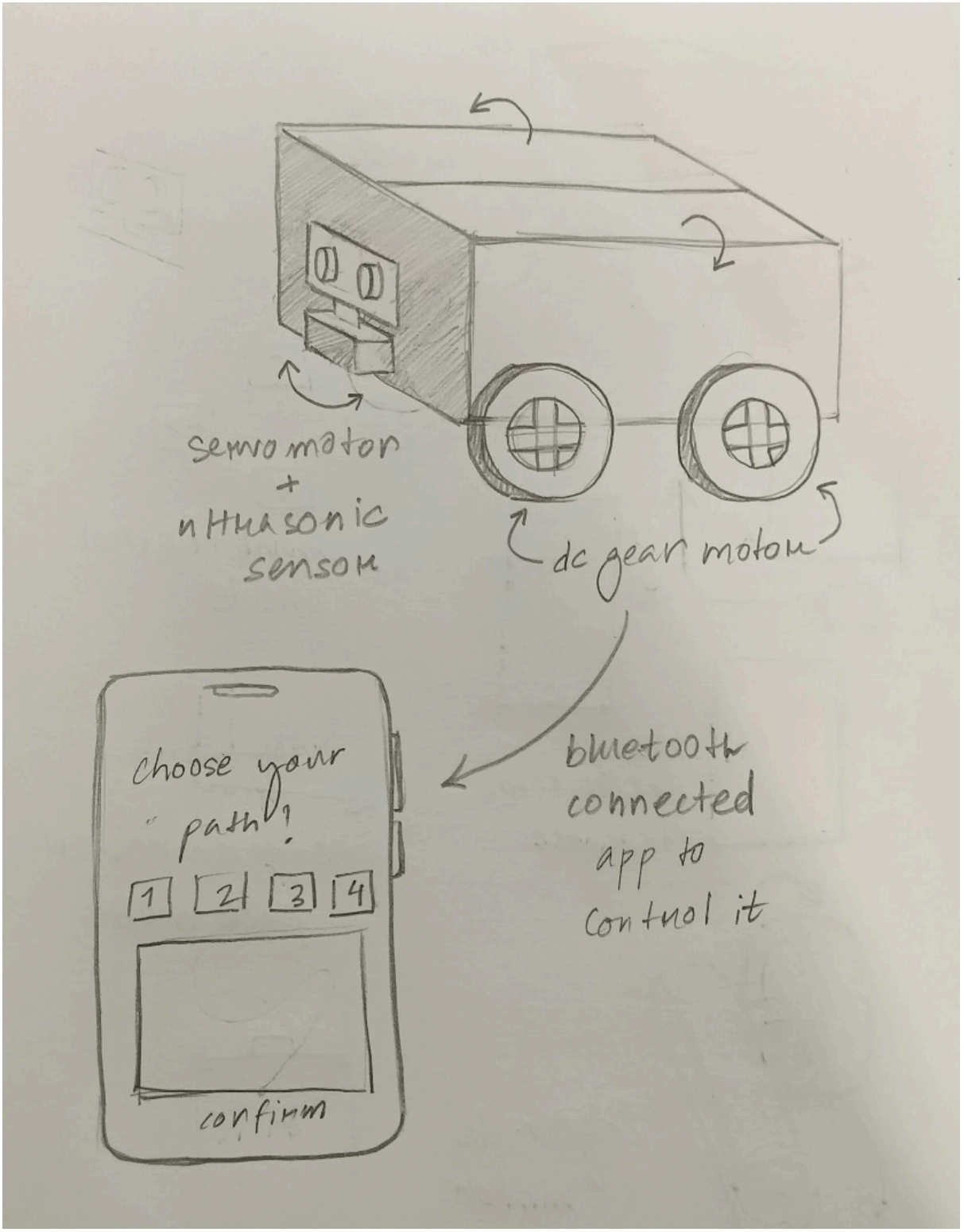
finalised.

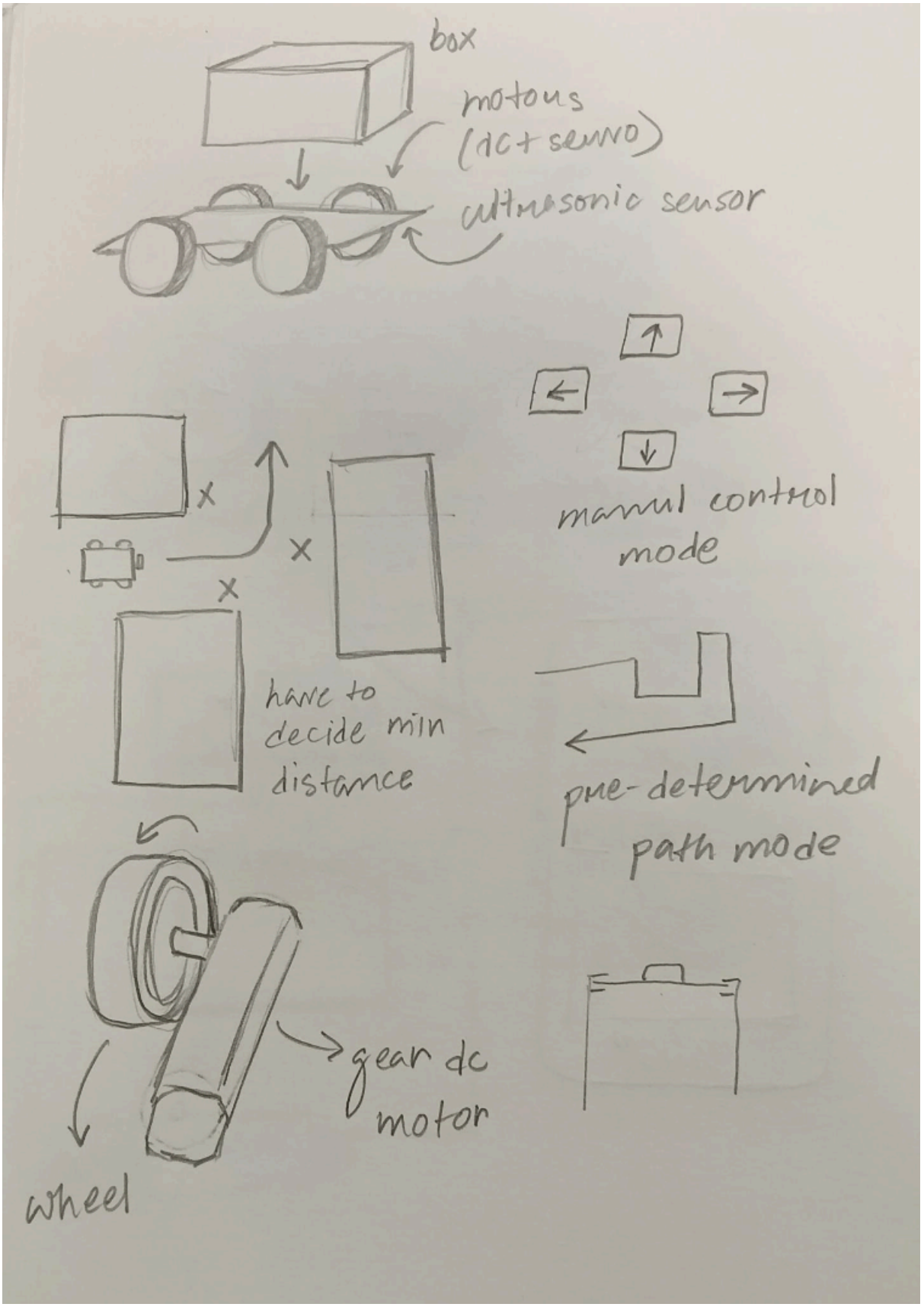
3. Laser cut all the individual panels for both boxes from their respective materials.

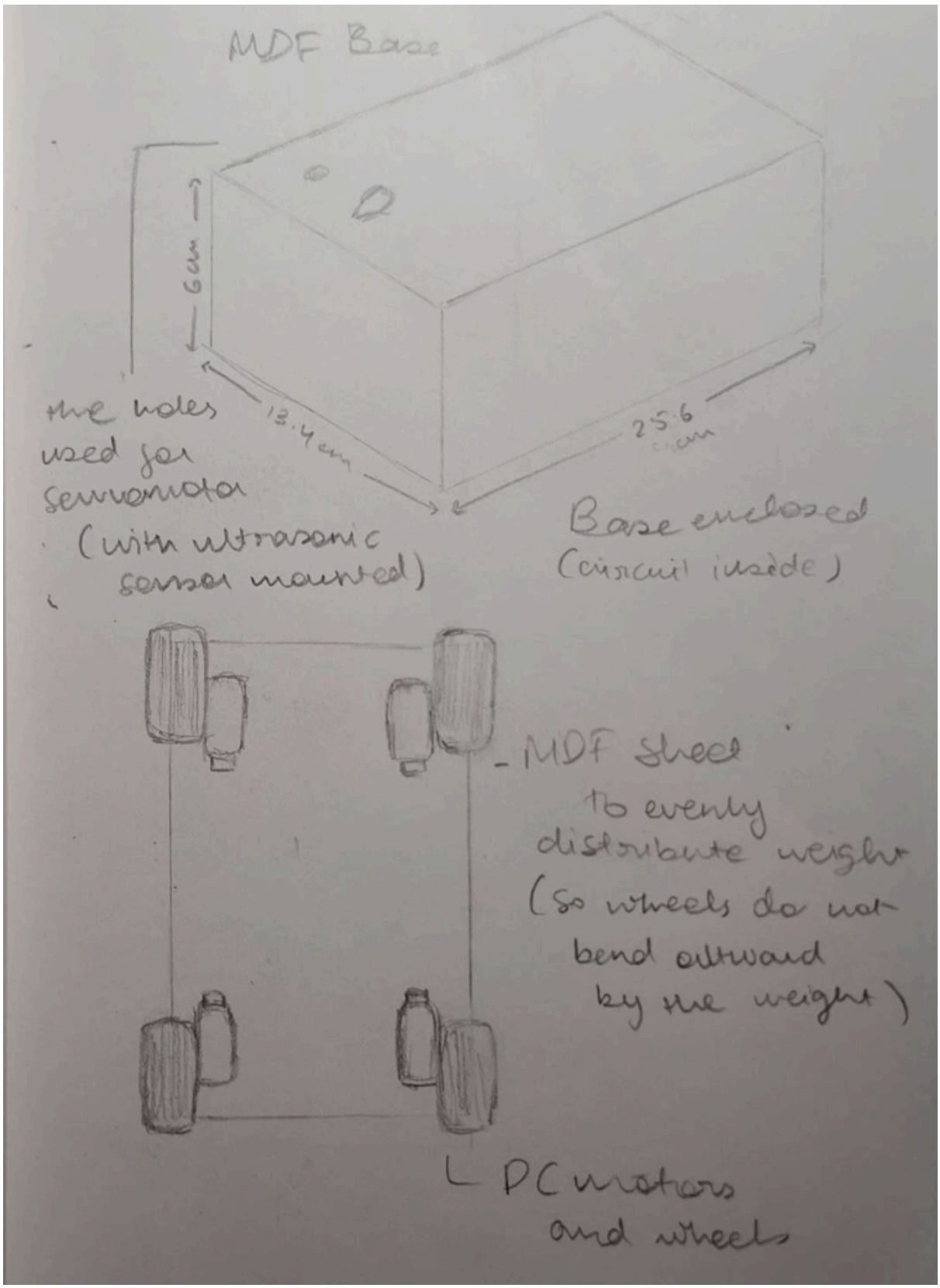
4. Temporarily attached both boxes as well as the chassis to test whether the motors could still drive all four wheels effectively under the additional weight of the structure.

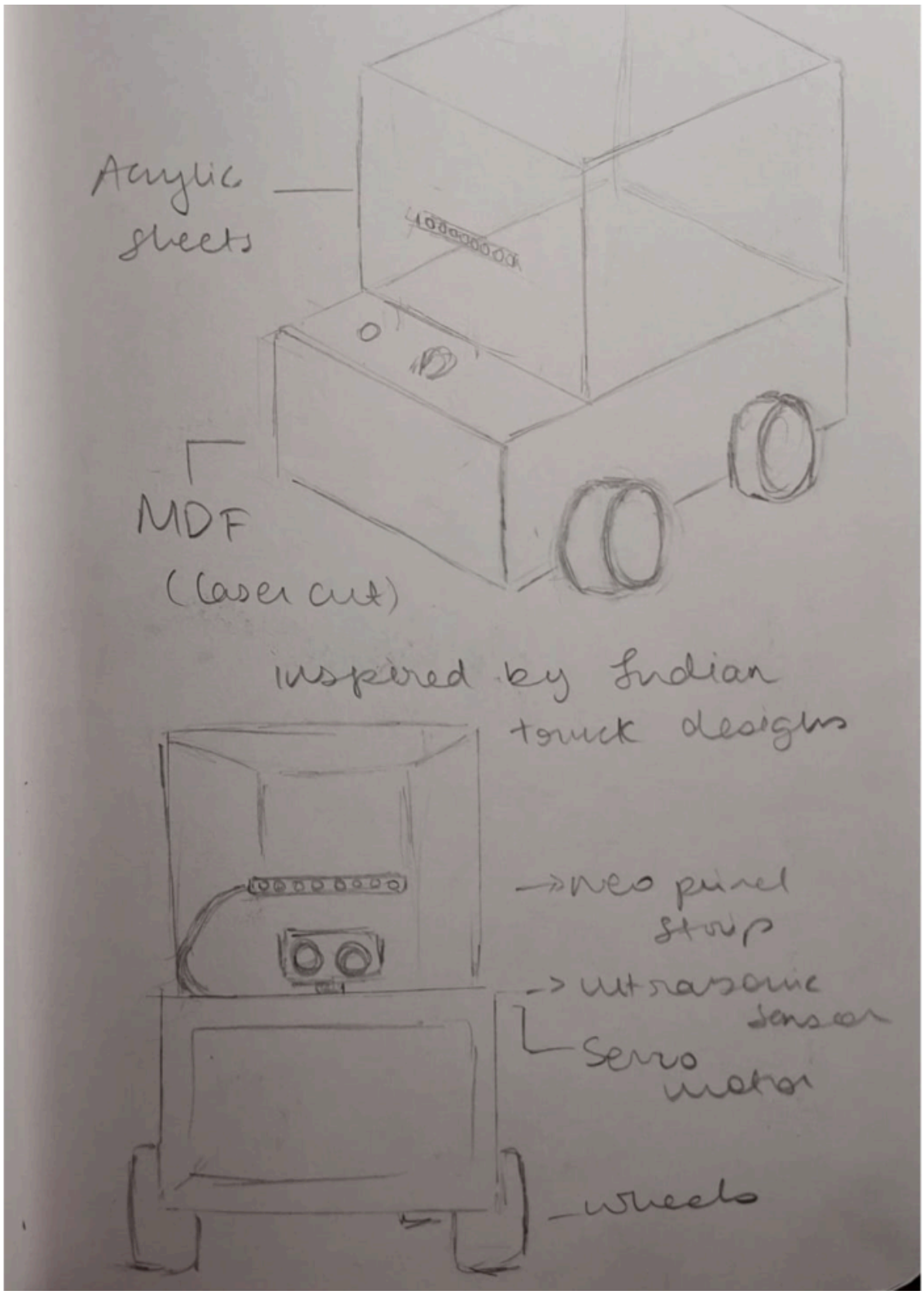
5. Cut and fitted a MDF sheet beneath the DC motors to evenly distribute the weight of both boxes across the wheel assembly, preventing the wheels from splaying outwards under the load.

6. After all testing and adjustments were completed, permanently secured the circuit inside the MDF enclosure, fixed all layers in place, and assembled the vehicle into its complete final form.


servo motor
+
ultrasonic
sensor
dc gear motor
bluetooth
connected
app to
control it
choose your
path!
1
2
3
4
confirm


box
motors
(dc + servo)
ultrasonic sensor
manual control
mode
have to
decide min
distance
pre-determined
path mode
gear dc
motor
wheel


MDF Base
6cm
13.4cm
25.6 cm
the holes used for servomotor (with ultrasonic sensor mounted)
Base enclosed (circuit inside)
MDF sheet to evenly distribute weight (so wheels do not bend outward by the weight)
DC motors and wheels


Acrylic sheets
MDF
(laser cut)
inspired by Indian truck designs
→ neo pixel strip
→ ultrasonic sensor
Servo motor
wheels